

Graph Learning for Cross-Subject, Cross-Population EEG Emotion Decoding and Model-Derived Spatial–Spectral Neural Signatures

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Abstract

Electroencephalography (EEG) provides a noninvasive means of capturing emotion-related neural dynamics, yet reliable EEG emotion decoding lacks models that can both generalize to unseen individuals and populations while preserving neural interpretability. To address these challenges, EmoDiPyraTrans is proposed as a development-regularized differential graph Transformer that models temporally ordered relative power spectral density graphs through adaptive graph recurrence, differential attention, and multiscale fusion. The framework was evaluated at three connected levels. First, cross-subject evaluations on SEED, FACED, MAHNOB-HCI, DEAP and DREAMER yielded participant-mean accuracies of 0.928, 0.645, 0.714, 0.617 and 0.671, respectively; the model ranked first among the evaluated methods for accuracy and positive-class F1 on all five datasets. Across seven ablation protocols, differential attention was the only component whose removal reduced both metrics in every case, whereas removing maximum mean discrepancy reduced accuracy throughout. Second, DEP-EEG distinguished within- from cross-population positive-versus-neutral decoding. Accuracy was 0.802 within healthy controls, 0.704 within participants with depression and 0.591 under healthy-to-depression transfer. Mixed-population development produced 0.581 accuracy and the highest positive-class F1 (0.498), indicating that greater population diversity alone did not remove the transfer gap. Third, channel- and frequency-resolved analyses on SEED identified a distributed frontal, temporal, central and parietal pattern, an alpha-centred low-to-mid-frequency preference and a six-channel subset that preserved near-full performance. This progression links broad cross-subject validation to an explicit clinical population-shift stress test and then to model-derived candidate spatial–spectral neural signatures. These signatures remain SEED-specific computational hypotheses rather than validated biomarkers or evidence of shared physiology across populations.

Keywords: EEG emotion decoding, cross-subject generalization, cross-population transfer, differential attention, distribution regularization, model-derived neural signatures

Electroencephalographic (EEG) emotion decoding seeks to infer affective states from non-invasive measurements of rapid neural activity. Its practical and scientific value depends on more than high accuracy in familiar participants: a useful decoder should generalize to unseen individuals, remain informative when development and evaluation populations differ, and expose the neural evidence supporting its decisions. These requirements are linked. EEG measurements vary with anatomy, skull conductivity and electrode contact [1, 2], while emotion engages distributed brain systems rather than a single locus [3]. A representation that removes individual variation at the expense of channel or frequency structure may transfer but cannot support spatial–spectral interpretation; conversely, an interpretable pattern confined to one cohort may reflect cohort-specific recording structure rather than transferable affective information.

Generalization therefore has at least two distinct levels. Cross-subject decoding asks whether affective valence can be

inferred under leave-one-subject-out (LOSO) evaluation without individual calibration. Cross-population decoding is a stricter test because cohort composition and clinical status can introduce shifts beyond participant-level variability. DEP-EEG provides a direct setting in which to separate these questions using recordings from healthy controls and participants with depression [4, 5]. Within-population evaluation establishes whether positive-versus-neutral emotion can be decoded separately in each population; healthy-to-depression transfer tests whether model parameters learned only from healthy controls remain informative in participants with depression; and mixed-population development tests whether adding population diversity improves generalization or introduces negative transfer. Comparing these settings makes population shift an explicit experimental quantity rather than a hidden source of performance loss. It does not, by itself, attribute that shift to depression-related neurophysiology, but it defines a clinically relevant boundary for calibration-free transfer.

Existing computational approaches address individual parts of this problem. Convolutional models capture temporal dynamics and spatial asymmetry, graph neural networks encode

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inter-channel relations, and Transformer variants model local and long-range dependencies [6, 7, 8, 9]. Domain-adversarial and multi-source alignment methods reduce distribution discrepancies across participants or sessions [10, 11, 12], whereas attention-based frameworks expose channel- or feature-level weighting [13]. However, predictive alignment and neural interpretation are commonly treated as separate objectives. Distribution control can obscure the sensor, frequency and temporal coordinates needed for attribution, while attention maps alone do not show whether selected channels or bands sustain prediction. The unresolved computational problem is therefore to control participant- and population-associated variation under strict cross-subject evaluation while preserving a structured representation that can be interrogated through independent spatial and spectral tests.

The Emotion Differential Attention Pyramid Transformer (EmoDiPyraTrans) addresses this problem using temporally ordered relative power spectral density (rPSD) graphs. These graphs preserve electrode identities, frequency-band attributes and temporal order, while parallel adaptive graph-recurrent branches and pyramid fusion capture channel dynamics at multiple scales [14]. Differential self- and cross-attention attenuate components shared between paired attention maps during channel refinement and graph aggregation [15]. Maximum mean discrepancy (MMD) and confidence-filtered pseudo-labels regularize two streams formed entirely from development data. A Differential Transformer then models long-range temporal dependencies. Because the representation retains its original spatial and spectral coordinates, the same framework can support both cross-subject prediction and channel- and frequency-resolved model interrogation.

The experiments followed a three-level progression. First, cross-subject evaluation on five public EEG emotion benchmarks, SEED [16], FACED [17], MAHNOB-HCI [18], DEAP [19] and DREAMER [20], tested generalization across unseen participants under heterogeneous cohort sizes, electrode montages and label definitions. Component and training-scheme ablations across these benchmarks and the within-population DEP-EEG protocols then separated consistently useful mechanisms from dataset-dependent ones. Second, four DEP-EEG protocols extended the evaluation from cross-subject classification to cross-population transfer: HC-LOSO, DEP-LOSO, direct healthy-to-depression transfer and mixed-population DEP-LOSO. This level distinguishes within-population learnability from the performance retained, or lost, when the population changes.

Third, SEED was used to determine whether a transferable decoder could generate a coherent model-derived spatial-spectral neural hypothesis. Its 62-channel montage, binary positive-negative contrast and seven-band rPSD representation enabled complementary analyses at both axes. Differential cross-attention contrasts and a trainable node mask provided independent spatial readouts; participant-level statistics and channel-pruning experiments tested their consistency and predictive sufficiency. Leave-one-band-out, single-band and band-combination experiments then quantified spectral dependence, while topographic comparisons tested whether changing

the spectral input reorganized the spatial solution. Together, the three experimental levels move from broad cross-subject validation, through a clinical cross-population stress test, to model-based neural feature identification. The resulting SEED signatures are framed as candidate computational hypotheses for independent physiological and reduced-montage validation, rather than universal biomarkers or evidence that the same neural pattern transfers across populations.

1. Results

1.1. Cross-subject decoding across public datasets

The LOSO results for SEED, MAHNOB-HCI, DEAP and DREAMER, together with the established cross-subject results for FACED, showed that EmoDiPyraTrans ranked first for participant-mean accuracy and positive-class F1 on every dataset (Table 1).

On SEED, EmoDiPyraTrans achieved 0.928 ± 0.055 accuracy and 0.928 ± 0.054 F1. These values exceeded those of MSGM, the strongest comparator for both metrics, by 9.4 and 7.8 percentage points, respectively. Their standard deviations were also lower than those of MSGM (0.114 for accuracy and 0.091 for F1). On FACED, accuracy reached 0.645 ± 0.048 , 1.1 percentage points above BiDANN, while F1 reached 0.768 ± 0.050 , 0.8 points above MSGM.

On MAHNOB-HCI, the model achieved 0.714 ± 0.079 accuracy and 0.611 ± 0.154 F1, exceeding the best baseline values by 12.7 and 21.5 percentage points. Accuracy on DEAP increased by 7.0 percentage points to 0.617 ± 0.073 , whereas F1 increased by 0.3 points to 0.679 ± 0.110 . On DREAMER, accuracy and F1 reached 0.671 ± 0.066 and 0.328 ± 0.273 , representing gains of 5.7 and 6.5 percentage points, respectively. The largest improvements therefore occurred on SEED and MAHNOB-HCI, while the remaining datasets extended the leading rankings across diverse recording and labelling conditions.

1.2. Emotion decoding within and across clinical populations

Positive-versus-neutral results on DEP-EEG were compared across HC-LOSO, DEP-LOSO, HC→DEP and mixed-population DEP-LOSO (Table 2).

Within healthy controls, EmoDiPyraTrans achieved 0.802 ± 0.130 accuracy and 0.797 ± 0.171 F1, exceeding DGCNN by 11.5 and 15.3 percentage points. Within participants with depression, accuracy and F1 reached 0.704 ± 0.094 and 0.709 ± 0.109 , exceeding TCN by 5.2 and 10.2 points. Under HC→DEP transfer, participant-mean accuracy and F1 were 0.591 and 0.598, respectively. Both values ranked first, exceeding TCN by 2.1 percentage points in accuracy and 13.8 points in F1. Mixed-population DEP-LOSO yielded a different metric profile. EmoDiPyraTrans achieved 0.581 ± 0.114 accuracy and the highest participant-mean positive-class F1 (0.498 ± 0.211). TCN led accuracy at 0.600 ± 0.100 , 1.9 percentage points higher, whereas EmoDiPyraTrans exceeded EmT in F1 by 1.6 points. Across the four protocols, the model therefore led both metrics for within-population and healthy-to-depression decoding, and led F1 under mixed-population development.

Table 1: Cross-subject emotion classification across five public EEG datasets.

Method	SEED		FACED		MAHNOB-HCI		DEAP		DREAMER	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
LSTM [21]	0.733 ± 0.158	0.670 ± 0.274	0.568 ± 0.063	0.700 ± 0.064	0.585 ± 0.126	0.341 ± 0.274	0.517 ± 0.099	0.570 ± 0.219	0.614 ± 0.106	0.125 ± 0.193
BiDANN [10]	0.794 ± 0.165	0.777 ± 0.159	0.634 ± 0.070	0.738 ± 0.064	—	—	—	—	—	—
DGCNN [7]	0.724 ± 0.145	0.619 ± 0.311	0.562 ± 0.045	0.697 ± 0.043	0.584 ± 0.104	0.376 ± 0.267	0.501 ± 0.094	0.495 ± 0.275	0.577 ± 0.116	0.169 ± 0.189
GCB-Net [22]	0.684 ± 0.172	0.517 ± 0.357	0.565 ± 0.052	0.685 ± 0.053	—	—	—	—	—	—
RGNN [23]	0.790 ± 0.148	0.802 ± 0.133	0.587 ± 0.050	0.722 ± 0.721	—	—	—	—	—	—
DMATN [24]	0.773 ± 0.155	0.772 ± 0.163	0.614 ± 0.049	0.683 ± 0.069	—	—	—	—	—	—
TSception [6]	0.662 ± 0.181	0.621 ± 0.283	0.619 ± 0.088	0.702 ± 0.237	0.580 ± 0.108	0.338 ± 0.219	0.546 ± 0.067	0.610 ± 0.226	0.597 ± 0.102	0.198 ± 0.163
TCN [25]	0.765 ± 0.140	0.737 ± 0.219	0.552 ± 0.035	0.673 ± 0.035	0.585 ± 0.128	0.344 ± 0.260	0.547 ± 0.085	0.564 ± 0.212	0.599 ± 0.094	0.263 ± 0.198
Conformer [8]	0.612 ± 0.127	0.529 ± 0.221	0.590 ± 0.035	0.720 ± 0.035	0.530 ± 0.087	0.140 ± 0.159	0.542 ± 0.094	0.676 ± 0.144	0.579 ± 0.106	0.147 ± 0.138
AMDET [26]	0.721 ± 0.168	0.649 ± 0.306	0.591 ± 0.043	0.726 ± 0.043	—	—	—	—	—	—
PGCN [27]	0.759 ± 0.184	0.741 ± 0.190	0.558 ± 0.078	0.665 ± 0.085	—	—	—	—	—	—
EmT [9]	0.802 ± 0.115	0.821 ± 0.093	0.608 ± 0.065	0.740 ± 0.058	0.587 ± 0.125	0.396 ± 0.248	0.514 ± 0.086	0.602 ± 0.183	0.599 ± 0.107	0.192 ± 0.202
MSGM [28]	0.834 ± 0.114	0.850 ± 0.091	0.632 ± 0.036	0.760 ± 0.037	—	—	—	—	—	—
EmoDiPyraTrans	0.928 ± 0.055	0.928 ± 0.054	0.645 ± 0.048	0.768 ± 0.050	0.714 ± 0.079	0.611 ± 0.154	0.617 ± 0.073	0.679 ± 0.110	0.671 ± 0.066	0.328 ± 0.273

Note. SEED and FACED baselines were taken from published comparisons in EmT [9] and MSGM [28]. MAHNOB-HCI, DEAP and DREAMER baselines were rerun under the corresponding protocols. Participant-specific metrics are reported as unweighted mean ± s.d. Acc, accuracy. F1 denotes positive-class F1 (positive emotion for SEED; high valence otherwise) and was set to 0 when no samples were predicted as positive. An em dash indicates an unavailable value; bold denotes the highest value per column.

Table 2: Cross-subject positive-versus-neutral classification on DEP-EEG.

Method	HC-LOSO		DEP-LOSO		HC→DEP		HC+DEP→DEP LOSO	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1
LSTM [21]	0.650 ± 0.124	0.562 ± 0.264	0.615 ± 0.114	0.576 ± 0.243	0.557	0.430	0.537 ± 0.092	0.345 ± 0.280
DGCNN [7]	0.687 ± 0.137	0.644 ± 0.248	0.577 ± 0.121	0.513 ± 0.263	0.528	0.385	0.532 ± 0.059	0.312 ± 0.260
TSception [6]	0.647 ± 0.125	0.589 ± 0.241	0.548 ± 0.094	0.455 ± 0.211	0.516	0.389	0.548 ± 0.069	0.409 ± 0.232
TCN [25]	0.636 ± 0.127	0.561 ± 0.254	0.652 ± 0.137	0.607 ± 0.269	0.570	0.460	0.600 ± 0.100	0.476 ± 0.261
Conformer [8]	0.610 ± 0.100	0.481 ± 0.249	0.532 ± 0.089	0.568 ± 0.119	0.516	0.148	0.574 ± 0.059	0.429 ± 0.192
EmT [9]	0.655 ± 0.122	0.625 ± 0.209	0.615 ± 0.145	0.566 ± 0.255	0.569	0.416	0.586 ± 0.116	0.482 ± 0.223
EmoDiPyraTrans	0.802 ± 0.130	0.797 ± 0.171	0.704 ± 0.094	0.709 ± 0.109	0.591	0.598	0.581 ± 0.114	0.498 ± 0.211

Note. HC, healthy control; DEP, participant with depression; LOSO, leave-one-subject-out; Acc, accuracy. Metrics were computed per participant. Values are mean ± s.d.; HC→DEP is the unweighted mean of 20 DEP participants. F1 is positive-emotion F1 and equals 0 when no samples are predicted as positive. Bold denotes the highest value per protocol–metric column.

1.3. Component and training-scheme ablations

Components of EmoDiPyraTrans were ablated on the five public datasets and under HC-LOSO and DEP-LOSO (Table 3). The cross-dataset results distinguished consistently beneficial components from those with dataset- or metric-specific effects.

Differential attention (DA) was the only component whose removal reduced both metrics in all seven protocols. On SEED, replacing DA with conventional multi-head attention reduced accuracy from 0.928 ± 0.055 to 0.842 ± 0.129 and F1 from 0.928 ± 0.054 to 0.830 ± 0.206 . Under HC-LOSO, accuracy and F1 fell from 0.802 and 0.797 to 0.668 and 0.583, respectively. Under DEP-LOSO, the corresponding values fell from 0.704 and 0.709 to 0.575 and 0.537. Across all protocols, accuracy decreased by 1.7–13.4 percentage points and F1 by 0.4–31.0 points.

Removing MMD reduced accuracy in every protocol, with decreases ranging from 0.6 percentage points on FACED to 10.9 points on SEED. Its effect on F1 was dataset specific. F1 decreased on SEED, MAHNOB-HCI and both DEP-EEG LOSO protocols, increased from 0.679 to 0.686 on DEAP and remained unchanged to three decimals on DREAMER. Thus, MMD consistently improved accuracy but had variable effects on precision–recall trade-offs.

Node enhancement (NE) and node aggregation (NA) showed

dataset-dependent effects. On SEED, removing NE or NA reduced accuracy to 0.891 and 0.921, respectively. By contrast, removing NA increased DEP-LOSO F1 from 0.709 to 0.737, while removing NE increased DREAMER accuracy from 0.671 to 0.685. Under HC-LOSO, removing NA, either alone or with NE, increased accuracy from 0.802 to 0.811. These reversals show that graph refinement was sensitive to channel montage and cohort composition.

Pseudo-label learning (PL), applied to confidence-filtered predictions from the target-validation stream, was similarly conditional. Removing PL reduced SEED accuracy and F1 from 0.928 to 0.925, but increased F1 in five protocols. The increases included DREAMER (0.328 to 0.377), FACED (0.768 to 0.778) and MAHNOB-HCI (0.611 to 0.616). On SEED, removing MMD alone yielded 0.819 accuracy, whereas removing both MMD and PL yielded 0.844. This interaction indicates that the contribution of PL depended on the representation shaped by distribution regularization.

Together, the ablations identified DA as the most consistently beneficial architectural component and MMD as the most consistent accuracy regularizer. NE, NA and PL provided dataset-dependent effects, making the complete architecture a balanced configuration across datasets and metrics.

Table 3: Croscomponent and training-scheme ablations of EmoDiPyraTrans.

Dataset / Protocol	Metric	Full	Component Ablation				Training-Scheme Ablation		
			w/o DA ^a	w/o NE	w/o NA	w/o NE & NA	w/o MMD	w/o PL	w/o MMD & PL
SEED	Acc	0.928 ± 0.055	0.842 ± 0.129	0.891 ± 0.100	0.921 ± 0.058	0.907 ± 0.078	0.819 ± 0.123	0.925 ± 0.051	0.844 ± 0.106
	F1	0.928 ± 0.054	0.830 ± 0.206	0.894 ± 0.096	0.922 ± 0.058	0.910 ± 0.072	0.827 ± 0.117	0.925 ± 0.051	0.830 ± 0.139
DEP-EEG (HC-LOSO)	Acc	0.802 ± 0.130	0.668 ± 0.149	0.810 ± 0.131	0.811 ± 0.120	0.811 ± 0.116	0.769 ± 0.134	0.795 ± 0.134	0.768 ± 0.130
	F1	0.797 ± 0.171	0.583 ± 0.279	0.788 ± 0.185	0.804 ± 0.129	0.809 ± 0.119	0.766 ± 0.157	0.795 ± 0.169	0.761 ± 0.163
DEP-EEG (DEP-LOSO)	Acc	0.704 ± 0.094	0.575 ± 0.127	0.674 ± 0.092	0.709 ± 0.104	0.691 ± 0.104	0.623 ± 0.080	0.705 ± 0.093	0.687 ± 0.093
	F1	0.709 ± 0.109	0.537 ± 0.228	0.665 ± 0.142	0.737 ± 0.086	0.698 ± 0.143	0.659 ± 0.117	0.713 ± 0.109	0.707 ± 0.097
DEAP	Acc	0.617 ± 0.073	0.597 ± 0.075	0.603 ± 0.071	0.613 ± 0.072	0.614 ± 0.069	0.609 ± 0.067	0.617 ± 0.072	0.615 ± 0.074
	F1	0.679 ± 0.110	0.588 ± 0.277	0.655 ± 0.161	0.666 ± 0.135	0.644 ± 0.177	0.686 ± 0.106	0.681 ± 0.114	0.685 ± 0.110
DREAMER	Acc	0.671 ± 0.066	0.607 ± 0.098	0.685 ± 0.068	0.668 ± 0.072	0.674 ± 0.073	0.652 ± 0.083	0.674 ± 0.068	0.669 ± 0.072
	F1	0.328 ± 0.273	0.055 ± 0.148	0.367 ± 0.270	0.343 ± 0.264	0.348 ± 0.247	0.328 ± 0.229	0.377 ± 0.259	0.285 ± 0.277
FACED	Acc	0.645 ± 0.048	0.628 ± 0.055	0.639 ± 0.053	0.642 ± 0.051	0.642 ± 0.052	0.639 ± 0.051	0.642 ± 0.051	0.638 ± 0.054
	F1	0.768 ± 0.050	0.764 ± 0.048	0.776 ± 0.041	0.775 ± 0.042	0.777 ± 0.040	0.768 ± 0.051	0.778 ± 0.038	0.770 ± 0.045
MAHNOB-HCI	Acc	0.714 ± 0.079	0.614 ± 0.091	0.693 ± 0.093	0.710 ± 0.096	0.709 ± 0.080	0.651 ± 0.129	0.713 ± 0.085	0.666 ± 0.106
	F1	0.611 ± 0.154	0.301 ± 0.294	0.578 ± 0.197	0.590 ± 0.205	0.606 ± 0.169	0.466 ± 0.263	0.616 ± 0.159	0.535 ± 0.220

Note. Acc, accuracy; w/o, without; Full, complete model; DA, differential attention; NE, node enhancement; NA, node aggregation. MMD, maximum mean discrepancy-based distribution alignment; PL, pseudo-label learning; HC, healthy control; DEP, participant with depression; LOSO, leave-one-subject-out. DEP-EEG uses within-population HC-LOSO and DEP-LOSO. Participant-specific metrics are reported as unweighted mean ± s.d. F1 is positive-emotion F1 for SEED and DEP-EEG and high-valence F1 otherwise; F1 equals 0 when no samples are predicted as positive. Bold denotes the largest value per dataset–metric row; tied maxima are all bold.

^a For w/o DA, standard multi-head mechanisms replaced differential self- and cross-attention in DiAttGE₁ and DiAttGE₂. They also replaced differential self-attention in the DiAttBlocks.

1.4. Hyperparameter sensitivity on SEED

Sensitivity analyses varied the graph hidden dimension, Differential Transformer depth and graph-convolution order on SEED while holding other settings fixed. The primary configuration used 32 hidden dimensions, eight Transformer layers and $K = 4$. Increasing the hidden dimension from 8 to 16 and 32 raised accuracy from 0.841 ± 0.143 to 0.901 ± 0.080 and 0.928 ± 0.055 , respectively. A dimension of 64 retained a mean accuracy of 0.928 (s.d., 0.073), but increased the parameter count from 841,962 to 3,215,610.

Depths of 2–8 layers produced accuracies of 0.925–0.928. Ten layers yielded the highest observed value (0.936 ± 0.054), whereas 12–16 layers yielded 0.921–0.930. The primary eight-layer setting remained within 0.8 percentage points of the maximum. For graph propagation, $K = 4$ produced 0.928 ± 0.055 , compared with 0.923 ± 0.077 at $K = 2$ and 0.916 at $K = 8$. The primary configuration therefore occupied a stable, high-accuracy region while limiting model depth and size.

1.5. SEED-derived spatial and spectral signature

1.5.1. Spatial organization and channel pruning

Fused attention, attention-contrast and node-mask maps were spatially non-uniform, with regionally structured between-participant variation (Fig. 1A–H). Combining the positive-minus-negative attention contrast with the node mask produced a distributed frontal, temporal, central and parietal pattern (Fig. 1G). False-discovery-rate-corrected channel-wise tests localized the strongest evidence to a subset of electrodes (Fig. 1I,J). Representative comparisons differed between conditions at Fpz, T7, AF3, P8, FC5, C6, P2 and C1, whereas PO6 and C5 provided non-significant reference examples (Fig. 1L).

Increasing the mask threshold reduced the retained montage through 62, 60, 54, 43, 31, 14, 9, 6 and 2 channels (Fig. 1K,M). Accuracy and F1 remained close to the full-montage values

with as few as six channels, but declined with two channels. The pruning profile thus identified a compact, distributed subset that preserved EmoDiPyraTrans performance on SEED.

1.5.2. Frequency dependence of the SEED signature

The full seven-band model achieved a mean accuracy of 0.928. Removing δ , θ , α , low β , β , high β or γ yielded 0.923, 0.928, 0.919, 0.919, 0.920, 0.930 and 0.930, respectively (Fig. 2A). Leave-one-band-out changes ranged from a 0.9-percentage-point decrease to a 0.2-point increase, indicating distributed contributions across the spectral representation.

Among single bands, α produced the highest mean accuracy (0.880), whereas β produced the lowest (0.597). The best two-band input, $\delta + \alpha$, reached 0.915, within 1.3 percentage points of the full model (Fig. 2B,C). Spatial organization also depended on spectral input. High β and γ maps were strongly correlated ($r = 0.91$), whereas low β correlated negatively with high β ($r = -0.78$) and γ ($r = -0.72$; Fig. 2D–G). Reduced multiband models formed distinct similarity patterns, with configuration-dependent central and centroparietal weighting (Fig. 2E–G). Together, these analyses identified an α -centred low-to-mid-frequency preference on SEED and showed that spectral input systematically reorganized spatial attention.

2. Discussion

EmoDiPyraTrans combines graph recurrence, differential attention and development-fold distribution regularization for calibration-free cross-subject and cross-population EEG emotion decoding. Across five public benchmarks, it ranked first among the evaluated methods for participant-mean accuracy and positive-class F1. All protocols isolated outer-test participants before model development. The results therefore connect predictive transfer across heterogeneous cohorts and pop-

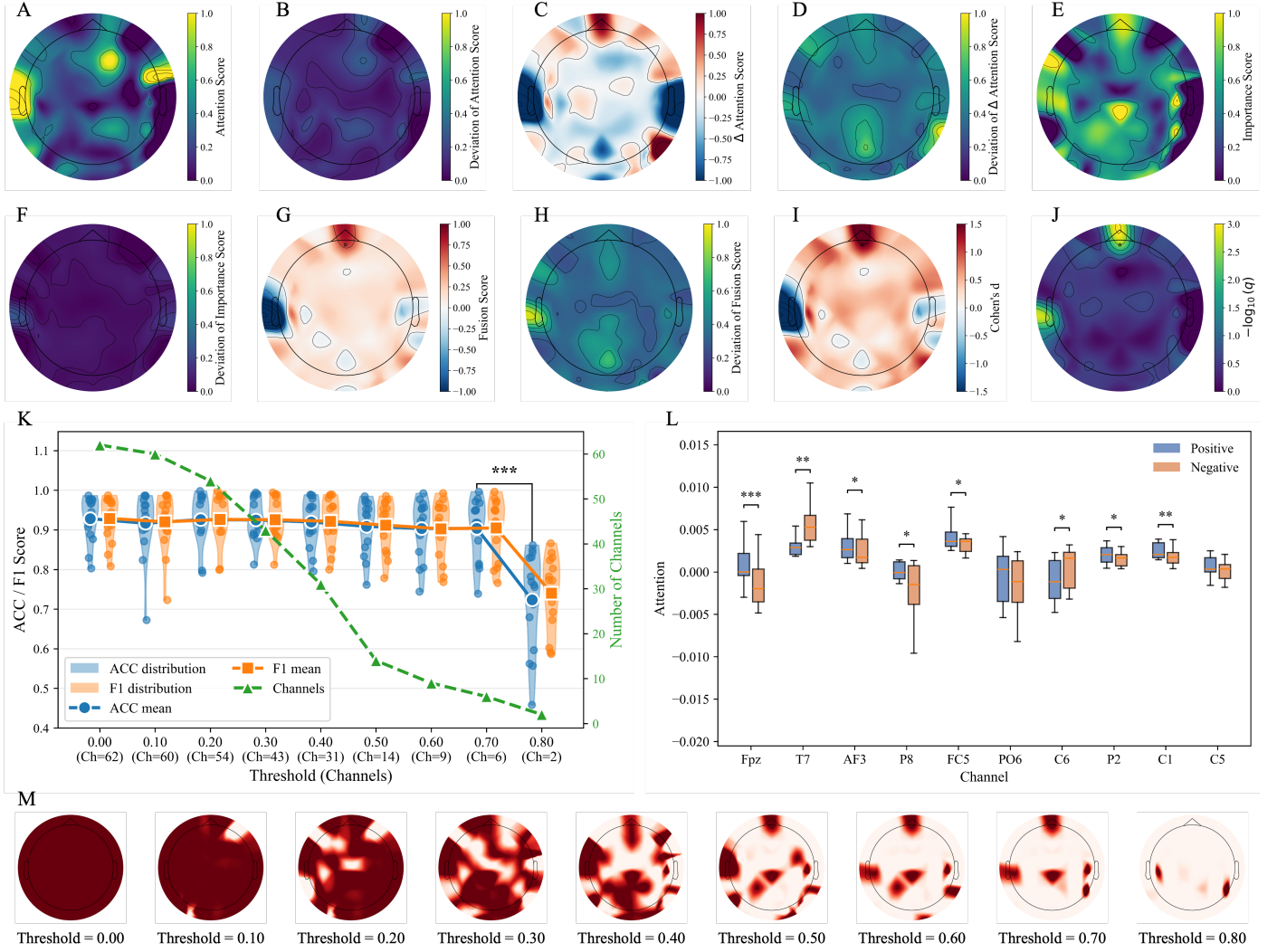


Figure 1: Model-derived spatial signature of binary emotion decoding on SEED. Multi-Head Differential Cross-Attention (MHDCA) weights from DiAttGE₁ and DiAttGE₂ were fused by element-wise multiplication. A trainable node mask provided a complementary estimate of channel importance. **A**, Group-mean fused cross-attention. **B**, Between-participant variance of **A**. **C**, Mean positive-minus-negative attention contrast. **D**, Between-participant variance of **C**. **E**, Mean node-mask score. **F**, Between-participant variance of **E**. **G**, Integrated mask $\times$ attention-contrast score. **H**, Between-participant variance of **G**. **I**, Paired Cohen's d_c . **J**, $-\log_{10}(q)$ after Benjamini-Hochberg correction of two-sided Wilcoxon signed-rank tests across channels. **K**, Participant-specific accuracy and positive-emotion F1 after retaining progressively fewer mask-selected channels and retraining under LOSO. Points denote test participants, lines denote means and the green curve gives the retained-channel count. **L**, Positive and negative distributions at representative electrodes. Boxes show medians and interquartile ranges; whiskers extend to 1.5 $\times$ IQR. **M**, Example topographies across pruning thresholds. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ for the comparisons shown.

ulations with channel- and frequency-resolved interrogation of the learned representation.

The public benchmarks also revealed dataset-dependent performance. Gains were largest on SEED and MAHNOB-HCI, whereas F1 improved by only 0.8 and 0.3 percentage points on FACED and DEAP. On the 14-channel DREAMER dataset, accuracy reached 0.671 but F1 was 0.328 ± 0.273 , indicating substantial metric-specific and between-participant variation. Although the rankings were consistent across recording conditions, smaller gains require more detailed uncertainty and class-wise analyses. Participant-level paired comparisons, balanced accuracy and recall could clarify the FACED and DEAP gains and the DREAMER metric divergence.

DEP-EEG extended this evaluation to positive-versus-neutral decoding in clinically distinct populations [4, 5].

Participant-mean accuracy/F1 reached 0.802/0.797 for HC-LOSO and 0.704/0.709 for DEP-LOSO. Under HC $\rightarrow$ DEP transfer, model parameters learned only from healthy controls yielded 0.591/0.598 across 20 participants with depression. Relative to HC-LOSO, accuracy and F1 decreased by 21.1 and 19.9 percentage points, respectively. The corresponding decreases relative to DEP-LOSO were 11.3 and 11.1 points. These contrasts quantify population-associated shift, but cohort size and clinical heterogeneity changed simultaneously. They therefore do not isolate depression-related neurophysiology as the cause of the performance difference.

Mixed-population DEP-LOSO provided a complementary test of population-diverse development for new participants with depression. EmoDiPyraTrans achieved 0.581 ± 0.114 accuracy and led positive-class F1 at 0.498 ± 0.211 , while TCN

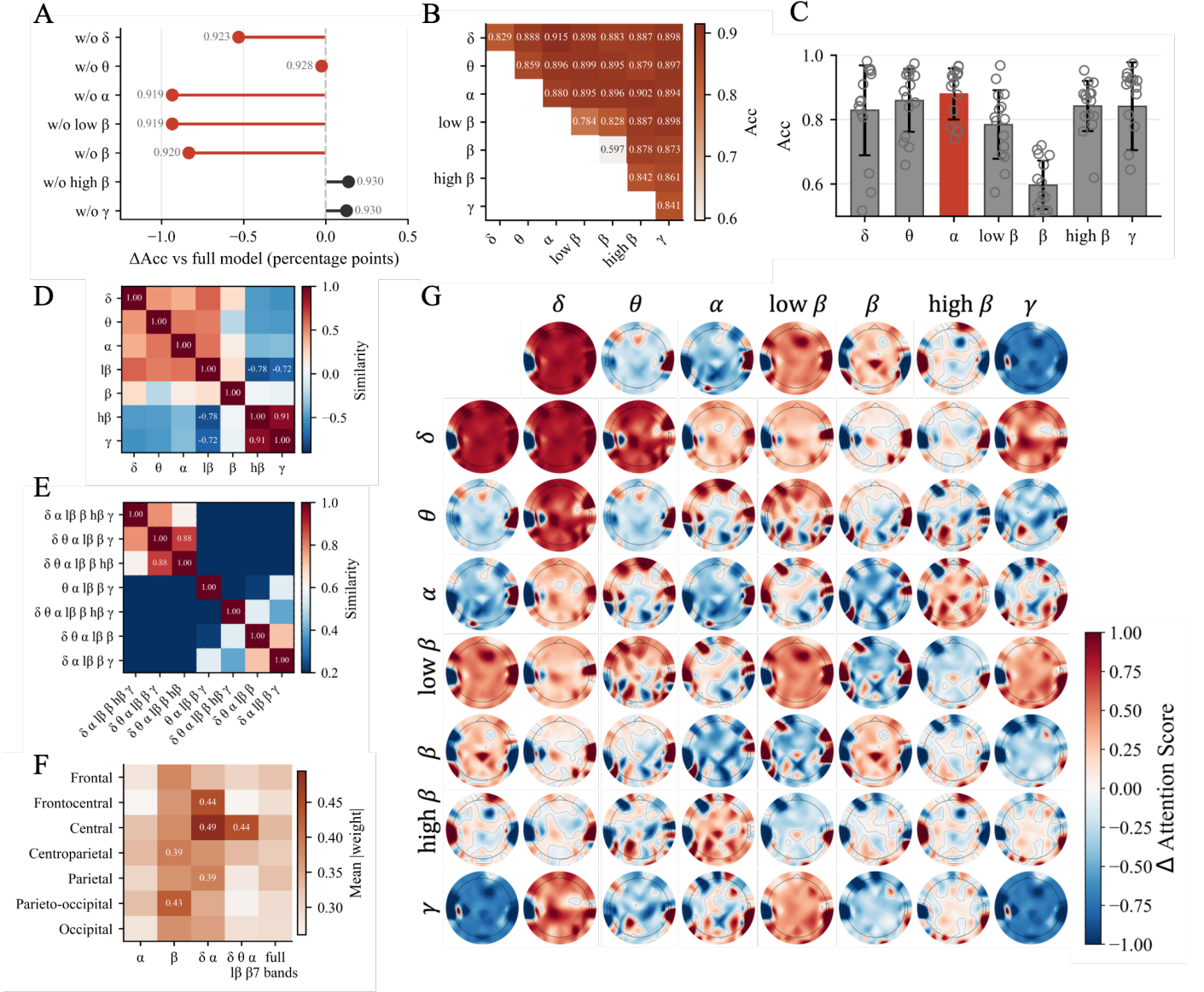


Figure 2: **Frequency dependence and spatial organization of the SEED-derived signature.** **A**, Accuracy of the full seven-band and leave-one-band-out models. **B**, Accuracy of single- and two-band inputs. **C**, Participant-level single-band accuracy ($n = 15$; bars, means; error bars, s.d.). **D**, Correlations between spatial attention-contrast maps from single-band models. **E**, Similarity among representative multiband models. **F**, Regional distribution of absolute spatial-attention weights for representative models. **G**, Group-mean positive-minus-negative attention-contrast maps across bands and representative combinations.

led accuracy at 0.600 ± 0.100 . Accuracy and F1 were 12.3 and 21.1 percentage points lower than under DEP-LOSO. They were also 1.0 and 10.0 points lower than under HC $\rightarrow$ DEP. Population imbalance and negative transfer are plausible contributors because cohort size, composition and internal partitioning changed together. Matched-size healthy-control subsets, population-balanced sampling and paired analyses over identical outer folds could distinguish these effects. One held-out participant had an F1 of 0, further motivating class-wise recall and macro-F1 reporting. These protocols address emotion decoding, not depression diagnosis, and do not validate the SEED signature across tasks.

The ablations separated broadly consistent components from dataset-dependent ones. Replacing differential attention re-

duced both metrics in all seven protocols, making it the most consistently beneficial architectural component. Removing MMD reduced accuracy in every protocol, although its effect on F1 varied. Pseudo-label learning was more conditional because its removal increased F1 in five protocols. Differential attention and MMD thus provided the most consistent contributions, while the complete objective balanced performance across datasets and metrics. Calibration-aware confidence selection or validation-based loss weighting may improve pseudo-label contributions [29].

The SEED analyses generated the spatial component of a model-derived candidate neural signature. Cross-attention contrasts, a complementary node mask and channel-wise statistics converged on a distributed frontal, temporal, central and

parietal pattern. Because mask optimization froze the encoder and classifier, the mask probed which channels sustained classification for a fixed predictor. Retraining with progressively smaller mask-selected montages preserved performance with as few as six channels, before performance declined with two. This finding supports subset sufficiency within SEED and motivates reduced-montage systems, but it does not establish neural necessity. The cohort-level consensus also requires independent validation using fold-specific selection with random and spatially matched controls. Source-resolved physiological measurements would be needed to test cortical localization [2, 1].

The spectral experiments further constrained this interpretation. Alpha was the best single band, and $\delta + \alpha$ was the best two-band pair; both approached the full representation. Removing individual bands changed mean accuracy by -0.9 to $+0.2$ percentage points, indicating distributed spectral contributions. Spectral selection also reorganized the spatial solution, as shown by opposing low- β and high-frequency maps and similar high- β and γ maps. Together, these findings support an alpha-centred low-to-mid-frequency model preference on SEED. Accounts involving sensory gating, attention or affect regulation remain hypotheses for independent physiological testing [16, 30].

Several limitations define the next validation stage. Dataset-specific montages, elicitation paradigms and label rules complicate claims of a shared physiological contrast. The model-derived SEED spatial-spectral neural signatures, derived from 15 participants, require replication across sessions, cohorts, tasks and hardware. Extensions of rPSD could incorporate phase, connectivity, cross-frequency coupling and event-related morphology. Repeated optimization would complement participant-level uncertainty estimates obtained from a fixed training seed. Prespecified nested or external validation is also needed to assess robustness beyond local sensitivity scans. Future DEP-EEG reports should document diagnostic criteria, medication, comorbidity, demographics, ethics and data-access conditions. Throughout these extensions, outer-test participants must remain isolated during both predictive and interpretation analyses.

In summary, EmoDiPyraTrans supported cross-subject EEG emotion decoding across heterogeneous datasets and quantified cross-population transfer under clinical population shifts. Its SEED analyses also defined model-derived candidate spatial-spectral neural signatures and a reduced channel subset. These outputs provide testable computational hypotheses for independent physiological and reduced-montage validation, rather than direct evidence of universal neural signatures.

3. Methods

3.1. EEG preprocessing and sequential rPSD graph construction

Continuous EEG recordings were represented as sequential relative power spectral density (rPSD) graphs for cross-subject decoding and spatial-spectral interpretation. The representation retained EEG channels as spatial entities, frequency-band

power as spectral attributes and subsegment order as temporal context.

For an EEG trial with C channels and L samples, the signal was denoted by $X \in \mathbb{R}^{C \times L}$. A sliding window of length l and hop s divided the trial into fixed-length segments $\tilde{X}_m \in \mathbb{R}^{C \times l}$. In this expression, $m = 1, \dots, M$ and $M = \lfloor (L - l)/s \rfloor + 1$. This segmentation avoided an assumption of stationarity across the complete trial and retained localized emotion-related EEG patterns.

Each segment $\tilde{X}_m$ was further divided into subsegments $\tilde{X}_{m,t} \in \mathbb{R}^{C \times l'}$ using a second window of length l' and hop s' . In this expression, $t = 1, \dots, T$ and $T = \lfloor (l - l')/s' \rfloor + 1$. Dataset-specific values of l , s , l' and s' are reported below. This second segmentation retained the temporal evolution of spatial-spectral EEG patterns within each segment.

Welch's method estimated the power spectral density independently for every channel in each subsegment $\tilde{X}_{m,t}$. The selected bands were delta (1–4 Hz), theta (4–8 Hz), alpha (8–12 Hz) and low beta (12–16 Hz). The remaining bands were beta (16–20 Hz), high beta (20–28 Hz) and gamma (30–45 Hz). Together, these seven bands retained an interpretable spectral axis for frequency-resolved analysis.

Let $\mathcal{B} = \{B_1, \dots, B_F\}$ denote the selected frequency bands, where $F = 7$. For channel c and band b , band power was computed as

$$P_{m,t,c,b} = \int_{\omega \in B_b} \text{PSD}_{m,t,c}(\omega) d\omega. \quad (1)$$

To reduce participant-dependent amplitude variation, absolute band power was normalized by the total power across all selected bands:

$$R_{m,t,c,b} = \frac{P_{m,t,c,b}}{\sum_{b'=1}^F P_{m,t,c,b'} + \epsilon}, \quad (2)$$

where ϵ is a small constant that ensures numerical stability. The resulting rPSD feature represented the relative distribution of oscillatory activity and improved comparability across participants.

Stacking rPSD values across channels and bands represented each subsegment as $G_{m,t} \in \mathbb{R}^{C \times F}$. This matrix defined the rPSD graph

$$\mathcal{G}_{m,t} = (\mathcal{V}, G_{m,t}), \quad (3)$$

where $\mathcal{V} = \{v_1, \dots, v_C\}$ denotes the EEG channel nodes. Each node corresponded to a scalp electrode and contained its rPSD vector across bands. This structure allowed discriminative responses to be mapped to electrode locations and spectral ranges.

Finally, the subsegment-level graphs were arranged in temporal order:

$$\mathcal{G}_m = \{\mathcal{G}_{m,1}, \mathcal{G}_{m,2}, \dots, \mathcal{G}_{m,T}\} \in \mathbb{R}^{C \times T \times F}. \quad (4)$$

The resulting sequential rPSD graph retained channel identities, frequency-band attributes and temporal order as the structured input for cross-subject EEG emotion decoding.

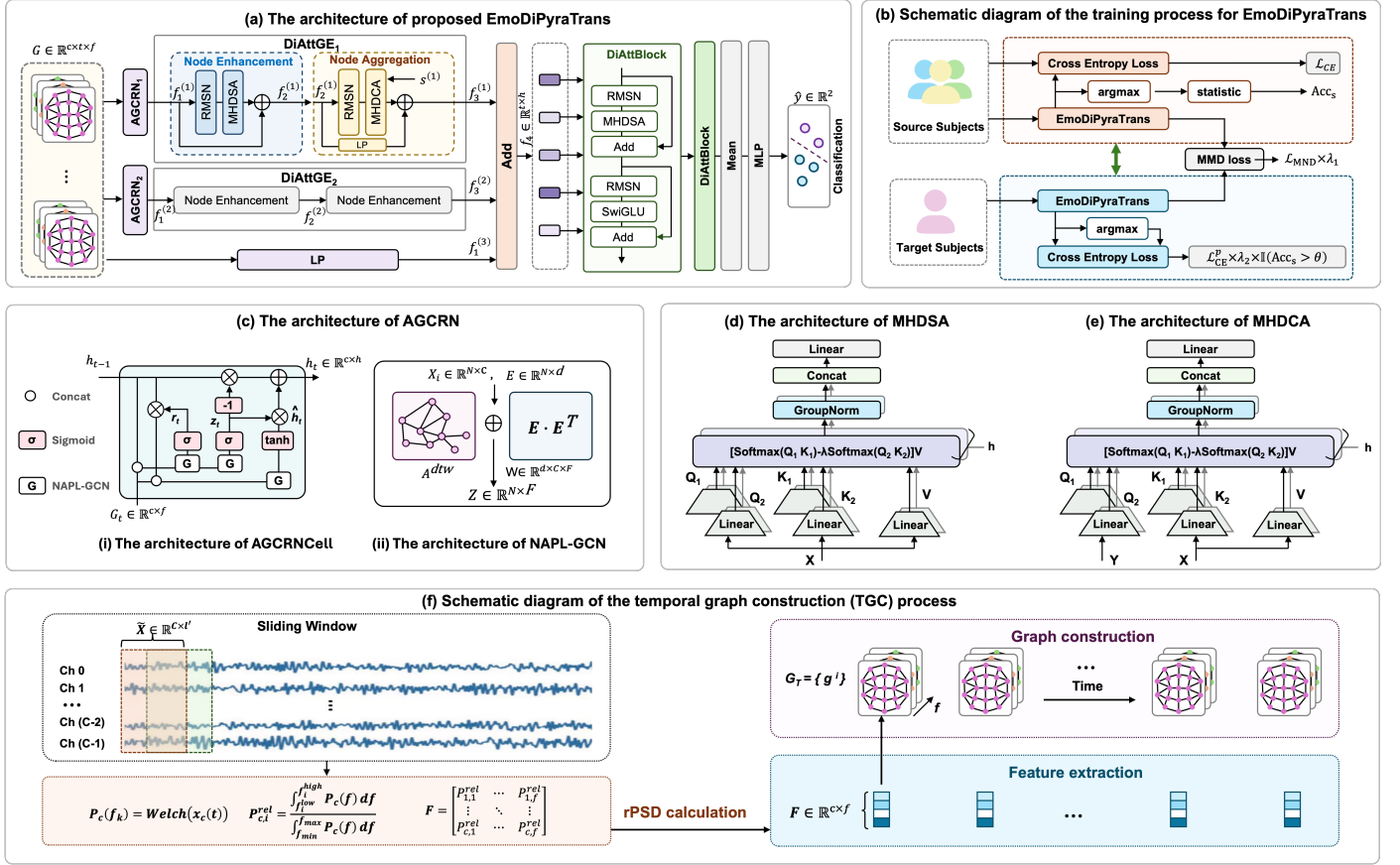


Figure 3: **Overview of EmoDiPyraTrans, its training scheme and temporal rPSD-graph construction.** **a**, Raw EEG is converted to relative power spectral density (rPSD) graphs $\mathcal{G} \in \mathbb{R}^{C \times T \times F}$ (channels $\times$ windows $\times$ frequency bands). Parallel one- and two-layer adaptive graph convolutional recurrent network (AGCRN) branches and a linear projection encode the graphs. Differential Attention-based Graph Enhancement modules (DiAttGE₁ and DiAttGE₂) refine these representations before pyramid fusion and classification by stacked Differential Attention Blocks. **b**, Dual-stream development procedure. After isolating outer-test participants, a protocol-specific partition assigns development samples to labelled source-training and target-validation streams. Within-participant protocols assign fixed class-stratified samples from each development participant to both streams. Source and target therefore describe stream roles rather than participant groups or populations. Source labels provide cross-entropy, while target-validation signals provide MMD and confidence-filtered pseudo-label losses. Their labels remain hidden from these objectives but provide validation accuracy for selecting the model parameters retained for evaluation. **c**, AGCRN, comprising an AGCRNCell and NAPL-GCN with adaptive adjacency from learnable node embeddings [14]. **d**, Multi-Head Differential Self-Attention (MHDSA) for feature refinement [15]. **e**, Multi-Head Differential Cross-Attention (MHDCA) for query-driven channel aggregation and its channel-attention output [15]. **f**, Temporal graph construction using sliding-window segmentation and Welch rPSD estimation.

3.2. Architecture of EmoDiPyraTrans

EmoDiPyraTrans extracted spatiotemporal features from time-varying rPSD graphs (Fig. 3a). The architecture comprised Adaptive Graph Convolutional Recurrent Networks (AGCRNs) [14], Differential Attention-based Graph Enhancement (DiAttGE), multiscale pyramid fusion and a Differential Transformer Encoder.

The input was a sequential rPSD graph $\mathcal{G} \in \mathbb{R}^{C \times T \times F}$, where C , T and F denote channels, temporal subsegments and frequency bands. Parallel one- and two-layer AGCRNs captured inter-channel relations and temporal dynamics (Fig. 3c). Their graph-spectral convolutions used adaptive adjacency matrices to model latent inter-channel dependencies. The branches yielded shallow and deep representations $f_1^{(1)}$ and $f_1^{(2)} \in \mathbb{R}^{C \times T \times H}$, where H denotes the embedding dimension. A concurrent linear projection produced the shallow representation $f_1^{(3)}$ for subsequent pyramid fusion.

DiAttGE processed $f_1^{(1)}$ and $f_1^{(2)}$ through node-enhancement

and node-aggregation stages. In DiAttGE₁, RMSNorm preceded Multi-Head Differential Self-Attention (MHDSA; Fig. 3d). A residual connection then combined the attention output with its input to produce $f_2^{(1)} \in \mathbb{R}^{C \times T \times H}$. Differential attention subtracted a scaled second attention map from the first, attenuating components shared by both maps [15].

RMSNorm and Multi-Head Differential Cross-Attention (MHDCA) subsequently aggregated the nodes (Fig. 3e). MHDCA related channel features $f_2^{(1)}$ to a learnable query $s^{(1)} \in \mathbb{R}^{T \times H}$. Adding a projected residual branch from $f_2^{(1)}$ to the MHDCA output produced $f_3^{(1)} \in \mathbb{R}^{T \times H}$. MHDCA weights quantified allocation across channels and were retained for spatial interpretation. DiAttGE₂ processed $f_1^{(2)}$ analogously to produce $f_3^{(2)}$.

Multiscale pyramid fusion integrated the graph-enhanced representations $f_3^{(1)}$ and $f_3^{(2)}$ with the shallow projection $f_1^{(3)}$. Element-wise addition merged the three representations into

$f_4 \in \mathbb{R}^{T \times H}$, thereby retaining information from different feature depths.

The Differential Transformer Encoder processed f_4 using stacked Differential Attention Blocks (DiAttBlocks). Each block combined MHDSA with feed-forward layers to model long-range temporal dependencies. Mean pooling aggregated the encoder output over time to produce $\bar{f}$. Two multilayer perceptron (MLP) layers then generated the classification output $\hat{y} \in \mathbb{R}^2$ for the target emotional state.

3.3. Cross-subject distribution regularization within the development fold

Test participants were isolated before model development in every outer evaluation fold. For the public datasets, HC-LOSO, DEP-LOSO and HC→DEP, each development participant was split once within emotion class using random seed 2022. The labelled source-training and target-validation streams received 80% and 20% of samples, respectively; FACED used a 90:10 split. Sample-index assignments were fixed and reused across models and training runs. Mixed-population DEP-LOSO instead used the joint population-by-class stratification described below.

At each iteration, the shared encoder received the source-training mini-batch $\{(\mathcal{G}_i^s, y_i^s)\}_{i=1}^{B_s}$. The corresponding target-validation mini-batch $\{\mathcal{G}_j^t\}_{j=1}^{B_t}$ provided a second input. Superscripts s and t denote stream roles, not healthy-control and depression populations. Target-validation labels were hidden from the MMD and pseudo-label objectives, but were used to select the model parameters retained for evaluation. The encoder $E_\theta(\cdot)$ produced $h_i^s = E_\theta(\mathcal{G}_i^s)$ and $h_j^t = E_\theta(\mathcal{G}_j^t)$. The classifier $C_\phi(\cdot)$ then produced softmax probabilities p_i^s and p_j^t .

Supervised discrimination was optimized using label-smoothed cross-entropy,

$$\mathcal{L}_{CE} = -\frac{1}{B_s} \sum_{i=1}^{B_s} \sum_{c=1}^{C_y} \tilde{y}_{i,c}^s \log p_{i,c}^s, \quad (5)$$

where C_y is the number of classes and $\tilde{y}^s$ is the one-hot target after label smoothing. Representation-level discrepancies between the two development streams were regularized by minimizing their squared maximum mean discrepancy:

$$\begin{aligned} \mathcal{L}_{MMD} = & \frac{1}{B_s^2} \sum_{i,i'=1}^{B_s} k(h_i^s, h_{i'}^s) + \frac{1}{B_t^2} \sum_{j,j'=1}^{B_t} k(h_j^t, h_{j'}^t) \\ & - \frac{2}{B_s B_t} \sum_{i=1}^{B_s} \sum_{j=1}^{B_t} k(h_i^s, h_j^t), \end{aligned} \quad (6)$$

where $k(\cdot, \cdot)$ is a positive-definite kernel.

Confidence-filtered pseudo-label learning was applied to the target-validation stream. Source mini-batch accuracy was defined as $a_s = B_s^{-1} \sum_i \mathbf{1}(\arg \max_c p_{i,c}^s = y_i^s)$. Pseudo-label learning was enabled when $a_s > \tau_a$. For target-validation sample j , $\hat{y}_j^t = \arg \max_c p_{j,c}^t$, $q_j = \max_c p_{j,c}^t$ and $m_j = \mathbf{1}(q_j > \tau_c)$. The loss was

$$\mathcal{L}_{PL} = -\frac{\mathbf{1}(a_s > \tau_a)}{\max(1, \sum_{j=1}^{B_t} m_j)} \sum_{j=1}^{B_t} m_j \log p_{j,\hat{y}_j^t}^t. \quad (7)$$

The total training objective was

$$\mathcal{L}_{total} = \mathcal{L}_{CE} + \lambda_{MMD} \mathcal{L}_{MMD} + \lambda_{PL} \mathcal{L}_{PL}. \quad (8)$$

This objective aligned source-training and target-validation representations during development. The parameters associated with the highest validation accuracy were retained.

3.4. Interpretability analysis and model-derived candidate signatures

SEED was used for interpretation because its positive-negative labels, 62-channel montage and seven-band rPSD representation supported channel- and frequency-resolved analyses. Preserved electrode and band identities allowed attention and masking variables to be mapped to scalp channels and spectral bands. Complementary spatial readouts and spectral perturbations together defined the model-derived candidate signatures.

3.4.1. Extraction and fusion of cross-attention-based spatial importance

Spatial interpretation used the Multi-Head Differential Cross-Attention (MHDCA) modules in DiAttGE₁ and DiAttGE₂. Each MHDCA module used a learnable query embedding to aggregate channel-wise features into a graph-level representation. The resulting cross-attention weights quantified the allocation of the model across EEG channels.

Indices denoted held-out participant n , sample i , DiAttGE branch $r \in \{1, 2\}$, time t , attention head h , query seed z and channel c . The corresponding MHDCA attention weight was $A_{n,i,t,h,z,c}^{(r)}$. Channel-level scores for each branch were obtained by averaging across temporal windows, attention heads and query seeds:

$$a_{n,i,c}^{(r)} = \frac{1}{THZ} \sum_{t=1}^T \sum_{h=1}^H \sum_{z=1}^Z A_{n,i,t,h,z,c}^{(r)}, \quad r \in \{1, 2\}. \quad (9)$$

The branch-specific scores were fused by element-wise multiplication, $a_{n,i,c} = a_{n,i,c}^{(1)} \cdot a_{n,i,c}^{(2)}$. This operation suppressed branch-specific fluctuations and emphasized channels selected at both graph-enhancement depths. For visualization, fused scores were min-max normalized across channels, with a small constant ϵ ensuring numerical stability.

Condition-specific attention maps were computed separately for positive and negative samples. Let $\mathcal{I}_n^+$ and $\mathcal{I}_n^-$ denote these sample sets for participant n . The participant-level attention difference at channel c was defined as

$$\Delta a_{n,c} = \frac{1}{|\mathcal{I}_n^+|} \sum_{i \in \mathcal{I}_n^+} \tilde{a}_{n,i,c} - \frac{1}{|\mathcal{I}_n^-|} \sum_{i \in \mathcal{I}_n^-} \tilde{a}_{n,i,c}. \quad (10)$$

Positive values indicated greater model attention for positive emotion, whereas negative values indicated greater attention for negative emotion.

3.4.2. Trainable node-mask prior for channel-level selection

Cross-attention weights directly measure model allocation, but may reflect both discriminative and compensatory behaviour. A trainable node mask was therefore introduced as a complementary estimate of channel importance. A participant-specific mask model was initialized from the trained EmoDiPyraTrans encoder and assigned a learnable vector $m_n \in \mathbb{R}^C$. Encoder and classifier parameters remained frozen during optimization, so only m_n was updated. The mask reweighted the input as $\mathcal{G}'_{n,i,t,c,b} = \sigma(m_{n,c})\mathcal{G}_{n,i,t,c,b}$, where $\sigma(\cdot)$ denotes the sigmoid function. Optimization preserved emotion classification while encouraging sparse and stable channel selection through the objective

$$\mathcal{L}_{\text{mask}} = \mathcal{L}_{\text{CE}} + \lambda_s \mathcal{L}_{\text{sparse}} + \lambda_e \mathcal{L}_{\text{entropy}} + \lambda_k \mathcal{L}_{\text{KL}}, \quad (11)$$

where $\mathcal{L}_{\text{CE}}$ evaluated classification through the frozen model and $\mathcal{L}_{\text{sparse}}$ penalized excessive mask activation. $\mathcal{L}_{\text{entropy}}$ encouraged near-binary decisions, while $\mathcal{L}_{\text{KL}}$ constrained mean activation towards the sparse target rate ρ . After optimization, mask scores were normalized within each participant.

The integrated spatial score was $B_{n,c} = \tilde{m}_{n,c} \cdot \Delta a_{n,c}$. Figure 1L used condition-specific scores $B_{n,c}^\pm = \tilde{m}_{n,c} |\mathcal{I}_n^\pm|^{-1} \sum_{i \in \mathcal{I}_n^\pm} \tilde{a}_{n,i,c}$. Thus, $B_{n,c} = B_{n,c}^+ - B_{n,c}^-$. Scores were rescaled within each participant for visualization. Group means and between-participant variability were calculated as the mean and s.d. across participants, respectively.

3.4.3. Statistical assessment and channel-pruning validation

For each channel, a two-sided Wilcoxon signed-rank test assessed whether $B_{n,c}$ differed from zero across participants. The Benjamini–Hochberg procedure corrected the resulting P values across channels. Statistical evidence was displayed as $-\log_{10}(q)$, and paired effect size was $d_z = \text{mean}(B_{n,c}) / \text{s.d.}(B_{n,c})$. Panel L compared $B_{n,c}^+$ and $B_{n,c}^-$ using the paired tests shown. These statistics quantified the consistency and magnitude of the model-derived contrast across participants.

Channel pruning was then performed. For each $\theta \in [0, 1]$, the global range of normalized mask scores defined a threshold. A cohort-level consensus set retained channels exceeding that threshold in every participant:

$$C_\theta = \{c \mid \tilde{m}_{n,c} \geq \tau_\theta, \forall n\}. \quad (12)$$

The model was retrained and evaluated with the retained channels under the same SEED LOSO protocol. Paired non-parametric tests compared accuracy and F1 across thresholds. This analysis quantified performance preservation within SEED as the montage became progressively smaller.

3.4.4. Frequency-band contribution and spectral signature analysis

Frequency-band ablation and combination analyses characterized spectral dependence on SEED. Let $\mathcal{B} = \{\delta, \theta, \alpha, \text{low } \beta, \beta, \text{high } \beta, \gamma\}$. For $\mathcal{S} \subseteq \mathcal{B}$, inputs were restricted to $\mathcal{G}^{\mathcal{S}} = \mathcal{G}_{::,b \in \mathcal{S}}$. EmoDiPyraTrans was then retrained on the same participant-level folds using only $\mathcal{G}^{\mathcal{S}}$.

Three complementary analyses were performed. First, leave-one-band-out analysis measured each band's contribution as $\Delta \text{Acc}_{\setminus b} = \text{Acc}_{\mathcal{B} \setminus \{b\}} - \text{Acc}_{\mathcal{B}}$. Negative values indicated that removing band b reduced performance. Second, single- and two-band models assessed standalone discrimination and pairwise complementarity, with $\text{Acc}_b = \text{Acc}_{\{b\}}$ and $\text{Acc}_{b_1, b_2} = \text{Acc}_{\{b_1, b_2\}}$. Third, reduced multiband models tested whether a compact low-to-mid-frequency subset preserved or improved performance relative to all seven bands.

For each configuration $\mathcal{S}$, MHDCA weights from DiAttGE₁ and DiAttGE₂ were extracted and fused as in the spatial analysis. This produced $a_{n,i,c}^{\mathcal{S}} = a_{n,i,c}^{(1,\mathcal{S})} \cdot a_{n,i,c}^{(2,\mathcal{S})}$. The corresponding condition-difference map was

$$\Delta a_c^{\mathcal{S}} = \frac{1}{N} \sum_{n=1}^N \left[\frac{1}{|\mathcal{I}_n^+|} \sum_{i \in \mathcal{I}_n^+} a_{n,i,c}^{\mathcal{S}} - \frac{1}{|\mathcal{I}_n^-|} \sum_{i \in \mathcal{I}_n^-} a_{n,i,c}^{\mathcal{S}} \right]. \quad (13)$$

This calculation produced a spatial-attention topography for each frequency-band configuration.

Pairwise topographic similarity was the Pearson correlation between condition-difference maps, $R_{S_1, S_2} = \text{corr}(\Delta a^{S_1}, \Delta a^{S_2})$. Electrodes were grouped into frontal, frontocentral, central, centroparietal, parietal, parieto-occipital and occipital regions. Within each region $\mathcal{R}$, $|\Delta a_c^{\mathcal{S}}|$ was averaged across channels. Together, frequency ablation, topographic similarity and regional summaries characterized how spectral inputs affected accuracy and the model-derived spatial solution.

3.5. Experimental setup

3.5.1. Datasets and preprocessing

EmoDiPyraTrans was evaluated on five public EEG emotion benchmarks (SEED, FACED, MAHNOB-HCI, DEAP and DREAMER) and the clinical DEP-EEG dataset. All tasks were binary and cross-subject. Each outer fold designated test participants before partitioning the remaining data for model development. The public datasets evaluated valence decoding. DEP-EEG evaluated positive-versus-neutral decoding within each population, HC→DEP transfer and mixed-population generalization to held-out participants with depression. SEED was also used for spatial and spectral interpretation.

SEED [16] contains EEG recordings from 15 participants who watched 15 Chinese film clips eliciting positive, negative and neutral emotions. The analysis used 62 EEG channels processed at 200 Hz. Neutral trials were excluded for binary valence classification, with positive and negative samples assigned to classes 1 and 0, respectively.

FACED [17] contains EEG recordings from 123 participants who watched 28 emotion-inducing video clips. The analysis used 32 channels sampled at 250 Hz. Processed EEG and affective ratings were obtained from the released data and corresponding annotation files. Valence scores greater than 3.0 defined the high-valence class; scores less than or equal to 3.0 defined the low-valence class.

For MAHNOB-HCI [18], the analysis used data from 27 participants, 32 EEG channels and a sampling rate of 128 Hz. Non-EEG channels were removed, and signals were average-referenced. Each recording was cropped by 30 s at both ends,

band-pass filtered at 0.3–45 Hz and notch-filtered at 50 Hz. Binary classification used the 1–9 SAM-like valence ratings with a threshold of 5. Samples at the threshold were discarded; scores above and below 5 defined the high- and low-valence classes, respectively.

For DEAP [19], the analysis used official preprocessed files from 32 participants, with 32 EEG channels sampled at 128 Hz. The first 3 s baseline period was removed. Valence scores above 5.0 defined the high-valence class, whereas scores at or below 5.0 defined the low-valence class. Because the released signals were already filtered at 4–45 Hz, rPSD comprised six bands: θ , α , low β , β , high β and γ . The δ band used in the other datasets was therefore omitted.

DREAMER [20] contains EEG recordings from 23 participants, each completing 18 audiovisual emotion-elicitation trials. The analysis used 14 channels sampled at 128 Hz and retained the final 60 s of each trial. Valence scores above 3.0 defined the high-valence class; scores at or below 3.0 defined the low-valence class.

DEP-EEG was obtained from the affective brain–computer–interface studies of Huang et al. and Guan et al. [4, 5]. It contains labelled EEG from 40 healthy controls and 20 participants with depression. Each participant viewed four neutral and four positive videos presented in randomised category order. Approximately 60 s of EEG were released per video, of which 50 s were retained. EEG was recorded from 30 scalp channels at 250 Hz, with the right mastoid A2 as the online reference. Released signals had undergone common-average re-referencing, high-pass filtering at 0.01 Hz and power-line-noise suppression. Positive and neutral epochs defined the two emotion classes and were converted to sequential rPSD graphs.

All trials were converted to sequential rPSD graphs as described above. Segmentation used 20-s windows with 80% overlap, corresponding to a 4-s hop. Each segment was subdivided with 75% overlap. Subsegments were 2 s for SEED and DREAMER, and 4 s for FACED, MAHNOB-HCI, DEAP and DEP-EEG. SEED, FACED, MAHNOB-HCI, DREAMER and DEP-EEG used bands of [1, 4], [4, 8], [8, 12], [12, 16], [16, 20], [20, 28] and [30, 45] Hz. DEAP used the same bands except [1, 4] Hz. The resulting input was $\mathcal{G} \in \mathbb{R}^{C \times T \times F}$, with C channels, T temporal subsegments and F frequency bands.

3.5.2. Evaluation protocol and baseline methods

All experiments were inductive with respect to the outer-test participants. SEED, MAHNOB-HCI, DEAP and DREAMER used leave-one-subject-out (LOSO) cross-validation, whereas FACED used leave-12-subjects-out evaluation. Each fold designated outer-test participants before forming and partitioning the development pool. The parameters associated with the highest validation accuracy were retained.

SEED, MAHNOB-HCI, DEAP and DREAMER held out one participant per fold. Within each remaining participant, fixed class-stratified samples were assigned to source-training and target-validation streams in an 8:2 ratio. FACED used ten outer folds, holding out 12 participants in each of the first nine folds and 15 in the final fold. Samples from every remaining FACED participant were divided 9:1. The parameter set with

the highest target-validation accuracy was used to evaluate each participant in the outer-test fold.

DEP-EEG used four protocols. HC-LOSO held out one of 40 healthy controls per fold and applied the fixed 8:2 split within every remaining participant. DEP-LOSO applied the same procedure to the 20 participants with depression. For HC→DEP, all 40 healthy controls contributed class-stratified samples to both development streams through fixed 8:2 splits. The resulting model was evaluated separately on each of the 20 participants with depression.

Mixed-population DEP-LOSO, denoted HC+DEP→DEP LOSO in Table 2, comprised 20 outer folds. Each fold isolated all 64 segments from one participant with depression before development. The remaining pool contained 3,776 segments from 40 healthy controls and 19 participants with depression. Using random seed 2022, joint stratification by population and emotion class divided these segments into training ($n = 3,020$) and validation ($n = 756$) subsets. This segment-level 8:2 split supplied the two development streams.

During optimization, target-validation signals supplied the MMD term and confidence-filtered pseudo-labels. Their ground-truth labels were used only to compute validation accuracy and select the retained model parameters. Accuracy (Acc) and positive-class F1 were computed from sample-level predictions for each test participant. Class 1 represented positive emotion for SEED and DEP-EEG, and high valence for the other datasets. Class 0 represented negative emotion for SEED, neutral emotion for DEP-EEG and low valence for the remaining datasets. F1 was set to zero when no samples were predicted as positive.

Reported means were unweighted averages of participant-specific metrics, rather than metrics calculated from pooled predictions. Standard deviations described variation across participants. For mixed-population DEP-LOSO, each fold contributed metrics from 64 test segments, and the reported s.d. used divisor $n = 20$. HC→DEP used a single parameter set for all 20 participants with depression and averaged their participant-specific values.

Comparators included recurrent, convolutional, graph-based, distribution-learning, Transformer and recent spatiotemporal EEG architectures. Tables 1 and 2 provide the original citation for each method. Rankings used participant-mean values reported or generated under the corresponding protocol.

3.5.3. Implementation details and hyperparameter settings

Experiments used PyTorch 2.1.1, CUDA 11.8 and Python 3.7 on Ubuntu 22.04 with one NVIDIA GeForce RTX 4080 GPU. The random seed was fixed at 2022.

The hidden graph dimension was 32, and the AGCRN node-embedding dimension was 10. The two AGCRN branches contained one and two recurrent graph-convolutional layers, respectively. Graph-convolution order was $K = 4$, and all datasets used eight attention heads. The Differential Transformer Encoder contained eight layers for all datasets except DREAMER, which used four. The primary SEED configuration was fixed before evaluation. Subsequent sensitivity scans did not inform the architecture or parameter set used in the primary evaluation.

For both MHDSA and MHDCA, the initial differential coefficient was $\lambda_{\text{init}} = 0.2$. MHDCA temperature was 0.5, and RMSNorm used $\epsilon = 10^{-5}$. In encoder layer l , the differential coefficient was initialized as $\lambda_{\text{init}}^{(l)} = 0.8 - 0.6 \exp[-0.3(l - 1)]$. The two-layer MLP classifier used a 32-to-16 projection, ReLU activation and batch normalization, followed by a 16-to-2 output layer.

Models were trained with mini-batches of 128 using AdamW and a learning rate of 3×10^{-4} . Dropout was 0.25, and label smoothing was 0.1. Training used at most 100 epochs for SEED and 50 epochs for all other datasets. Early-stopping patience was ten epochs.

The MMD and pseudo-label loss weights were both 0.5. MMD used a radial basis function kernel with five Gaussian kernels, a multiplier of 2.0 and adaptive bandwidth from pairwise feature distances. Pseudo-label learning was activated when source mini-batch accuracy exceeded $\tau_a = 0.8$. It retained target-validation predictions with a maximum softmax confidence above 0.9.

For the SEED node-mask analysis, sparsity, entropy and Kullback–Leibler-divergence weights were 5.0, 0.1 and 1.0, respectively. The sparse target activation rate was $\rho = 0.05$.

Data availability

SEED, FACED, MAHNOB-HCI, DEAP and DREAMER are third-party datasets available from their original providers. Access routes and terms are described in the corresponding dataset publications [16, 17, 18, 19, 20]. DEP-EEG is third-party human-participant EEG controlled by the investigators of the original affective brain–computer-interface studies [4, 5].

Code availability

Code for EmoDiPyraTrans and the associated analyses is available at <https://github.com/hdy6438/EmoDiPyraTrans>.

Ethics statement

This study involved secondary analysis of previously collected EEG datasets and therefore required no additional ethics approval.

Acknowledgements

This research was funded by the Natural Science Foundation of Chongqing (CSTB2025NSCQ-JM002, CSTB2025NSCQ-GPX0794 and CSTB2024NSCQ-MSX0118). Further support came from the Scientific and Technological Research Program of the Chongqing Education Commission (KJZD-K202303103, KJZD-K202501107 and KJQN202501104). The Chongqing Municipal Key Project for Technology Innovation and Application Development provided grants CSTB2024TIAD-KPX0042 and CSTB2025TIAD-KPX0002. The Hong Kong Polytechnic

University provided an internal grant (P0048377), a Departmental Collaborative Research Fund grant (P0056428) and a Collaborative Research with World-leading Research Groups Fund grant (P0058097). The Research Grants Council Collaborative Research Fund provided grant C5033-24G.

Author contributions

D.H.: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization and writing the original draft. B.J.: conceptualization, project administration, funding acquisition, supervision, and manuscript review and editing. X.W.: visualization, validation, and manuscript review and editing. Y.Z.: funding acquisition and manuscript review and editing. H.Y.: validation and manuscript review and editing. W.T.S.: funding acquisition and manuscript review and editing. N.W.: conceptualization, project administration, funding acquisition, supervision, validation, and manuscript review and editing.

Competing interests

All authors declare no competing interests.

References

- [1] M. Antonakakis, S. Schrader, Ü. Aydin, A. Khan, J. Gross, M. Zervakis, S. Rampp, C. H. Wolters, Inter-subject variability of skull conductivity and thickness in calibrated realistic head models, *Neuroimage* 223 (2020) 117353.
- [2] E. S. Kappenman, S. J. Luck, The effects of electrode impedance on data quality and statistical significance in erp recordings, *Psychophysiology* 47 (5) (2010) 888–904.
- [3] G. Mohammadi, D. Van De Ville, P. Vuilleumier, Brain networks subserving functional core processes of emotions identified with componential modeling, *Cerebral Cortex* 33 (12) (2023) 7993–8010.
- [4] W. Huang, W. Wu, M. V. Lucas, H. Huang, Z. Wen, Y. Li, Neurofeedback training with an electroencephalogram-based brain-computer interface enhances emotion regulation, *IEEE Transactions on Affective Computing* 14 (2) (2021) 998–1011.
- [5] Z. Guan, X. Zhang, W. Huang, K. Li, D. Chen, W. Li, J. Sun, L. Chen, Y. Mao, H. Sun, et al., A method for detecting depression in adolescence based on an affective brain-computer interface and resting-state electroencephalogram signals: Z. guan et al.: A method for detecting depression in adolescence based on an affective brain, *Neuroscience Bulletin* 41 (3) (2025) 434–448.
- [6] Y. Ding, N. Robinson, S. Zhang, Q. Zeng, C. Guan, Tsception: Capturing temporal dynamics and spatial asymmetry from eeg for emotion recognition, *IEEE Transactions on Affective Computing* 14 (3) (2022) 2238–2250.

- [7] T. Song, W. Zheng, P. Song, Z. Cui, Eeg emotion recognition using dynamical graph convolutional neural networks, *IEEE Transactions on Affective Computing* 11 (3) (2018) 532–541.
- [8] Y. Song, Q. Zheng, B. Liu, X. Gao, Eeg conformer: Convolutional transformer for eeg decoding and visualization, *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 31 (2022) 710–719.
- [9] Y. Ding, C. Tong, S. Zhang, M. Jiang, Y. Li, K. J. Lim, C. Guan, Emt: A novel transformer for generalized cross-subject eeg emotion recognition, *IEEE Transactions on Neural Networks and Learning Systems* (2025).
- [10] Y. Li, W. Zheng, Y. Zong, Z. Cui, T. Zhang, X. Zhou, A bi-hemisphere domain adversarial neural network model for eeg emotion recognition, *IEEE Transactions on Affective Computing* 12 (2) (2018) 494–504.
- [11] H. Chen, M. Jin, Z. Li, C. Fan, J. Li, H. He, Ms-mds: Multisource marginal distribution adaptation for cross-subject and cross-session eeg emotion recognition, *Frontiers in Neuroscience* 15 (2021) 778488.
- [12] J. Wang, X. Ning, W. Xu, Y. Li, Z. Jia, Y. Lin, Multi-source selective graph domain adaptation network for cross-subject eeg emotion recognition, *Neural Networks* 180 (2024) 106742.
- [13] R. Liu, Y. Chao, X. Ma, X. Sha, L. Sun, S. Li, S. Chang, Ertnet: an interpretable transformer-based framework for eeg emotion recognition, *Frontiers in Neuroscience* 18 (2024) 1320645.
- [14] L. Bai, L. Yao, C. Li, X. Wang, C. Wang, Adaptive graph convolutional recurrent network for traffic forecasting, *Advances in neural information processing systems* 33 (2020) 17804–17815.
- [15] T. Ye, L. Dong, Y. Xia, Y. Sun, Y. Zhu, G. Huang, F. Wei, Differential transformer, in: *International Conference on Learning Representations*, Vol. 2025, 2025, pp. 144–164.
- [16] W.-L. Zheng, B.-L. Lu, Investigating critical frequency bands and channels for eeg-based emotion recognition with deep neural networks, *IEEE Transactions on Autonomous Mental Development* 7 (3) (2015) 162–175.
- [17] J. Chen, X. Wang, C. Huang, X. Hu, X. Shen, D. Zhang, A large finer-grained affective computing eeg dataset, *Scientific Data* 10 (1) (2023) 740.
- [18] M. Soleymani, J. Lichtenauer, T. Pun, M. Pantic, A multi-modal database for affect recognition and implicit tagging, *IEEE Transactions on Affective Computing* 3 (1) (2011) 42–55.
- [19] S. Koelstra, C. Muhl, M. Soleymani, J.-S. Lee, A. Yazdani, T. Ebrahimi, T. Pun, A. Nijholt, I. Patras, Deap: A database for emotion analysis; using physiological signals, *IEEE Transactions on Affective Computing* 3 (1) (2011) 18–31.
- [20] S. Katsigiannis, N. Ramzan, Dreamer: A database for emotion recognition through eeg and ecg signals from wireless low-cost off-the-shelf devices, *IEEE Journal of Biomedical and Health Informatics* 22 (1) (2017) 98–107.
- [21] M. Soleymani, S. Asghari-Esfeden, Y. Fu, M. Pantic, Analysis of eeg signals and facial expressions for continuous emotion detection, *IEEE Transactions on Affective Computing* 7 (1) (2015) 17–28.
- [22] T. Zhang, X. Wang, X. Xu, C. P. Chen, Gcb-net: Graph convolutional broad network and its application in emotion recognition, *IEEE Transactions on Affective Computing* 13 (1) (2019) 379–388.
- [23] P. Zhong, D. Wang, C. Miao, Eeg-based emotion recognition using regularized graph neural networks, *IEEE Transactions on Affective Computing* 13 (3) (2020) 1290–1301.
- [24] F. Wang, W. Zhang, Z. Xu, J. Ping, H. Chu, A deep multi-source adaptation transfer network for cross-subject electroencephalogram emotion recognition, *Neural Computing and Applications* 33 (15) (2021) 9061–9073.
- [25] S. Zhang, C. Tang, C. Guan, Visual-to-eeg cross-modal knowledge distillation for continuous emotion recognition, *Pattern Recognition* 130 (2022) 108833.
- [26] Y. Xu, Y. Du, L. Li, H. Lai, J. Zou, T. Zhou, L. Xiao, L. Liu, P. Ma, Amdet: Attention based multiple dimensions eeg transformer for emotion recognition, *IEEE Transactions on Affective Computing* 15 (3) (2023) 1067–1077.
- [27] M. Jin, C. Du, H. He, T. Cai, J. Li, Pgc: Pyramidal graph convolutional network for eeg emotion recognition, *IEEE Transactions on Multimedia* 26 (2024) 9070–9082.
- [28] H. Liu, Y. Gong, Z. Yan, Z. Zhuang, J. Lu, Msgm: a multi-scale spatiotemporal graph mamba for eeg emotion recognition, *Frontiers in Neuroscience* 20 (2026) 1665145.
- [29] E. Arazo, D. Ortego, P. Albert, N. E. O’Connor, K. McGuinness, Pseudo-labeling and confirmation bias in deep semi-supervised learning, in: *2020 International Joint Conference on Neural Networks (IJCNN)*, IEEE, 2020, pp. 1–8.
- [30] L. Luther, J. M. Horschig, J. M. van Peer, K. Roelofs, O. Jensen, M. A. Hagenaars, Oscillatory brain responses to emotional stimuli are effects related to events rather than states, *Frontiers in Human Neuroscience* 16 (2023) 868549.